

Addressing Modes

Introduction to Assembly Programming and Addressing Modes

Addressing Modes

- Addressing mode (AM) is concerned with how data is **accessed**, not the way data is processed.
- The correct AM allows the CPU to identify the actual operand or the address location where operand is stored.
- The ARM processor instruction set architecture supports many different addressing modes.
 - Register direct
 - Immediate data
 - Register indirect
 - Register indirect with offset
 - Register indirect with index register
 - Pre and post auto-indexing

Learn More: Google “addressing modes”

MOV and LDR Instructions

Which of the following instructions **will not** result in
 $R1 = 0x00000200$ given the shown initial states

A) LDR R1, R0

Answer is A

B) MOV R1, #0x200

C) LDR R1, [R0]

D) None of the above

R0 0x00000200

R1 0x12345678

Initial State

Address Memory

0x200 0x00

0x201 0x02

0x202 0x00

0x203 0x00

0x204

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Register Indirect with Base & Offset

Given the initial states below, what is the value in R0 after executing the shown code segment.

A) 0x12345678

B) 0x00000100

C) 0x33221100

D) 0xAABBCCDD

```
MOV    R0, R1
MOV    R0, #0x100
LDR    R0, [R0, #4]
```

Answer is D

R0 0x00000000

R1 0x12345678

Initial State

Address Memory

0x100	0x00
0x101	0x11
0x102	0x22
0x103	0x33
0x104	0xDD
0x105	0xCC
0x106	0xBB
0x107	0xAA
0x108	

Immediate value

Which of the following instructions has an invalid immediate value

A) MOV R0, #0x100

B) MOV R0, #0x110

C) MOV R0, #0x102

D) MOV R0, #0x104

Answer is C

Immediate value

Which of the following instructions has an invalid immediate value

A) MOV R0, #256

Answer is B

B) MOV R0, #258

C) MOV R0, #0x258

D) MOV R0, #0x3F0

Register Indirect with Base & Offset

Given the initial states below, Which instruction will load 0x102 to R1 with the least number of clock cycles

A) MOV R1, #0x102

B) MOV R1, R0

C) LDR R0, [R2]

D) LDR R0, [R2, #-2]

Answer is B

R0 0x00000102

R1 0x12345678

Initial State

Address Memory

0x100	0x02
0x101	0x01
0x102	0x00
0x103	0x00
0x104	0x00
0x105	0x00
0x106	0x00
0x107	0x00
0x108	

Addressing modes

Which of the following instructions **will not** make $R2 = 0x00001100$ after execution?

A) `LDR R2, [R0, R1]`

B) `LDR R2, [R0, #1]`

C) `MOV R2, #0x1100`

D) None of the above

Answer is B

R0 `0x00001100`

R1 `0x00000004`

Initial State

Address Memory

0x1100	0x00
0x1101	0x00
0x1102	0x11
0x1103	0x00
0x1104	0x00
0x1105	0x11
0x1106	0x00
0x1107	0x00
0x1108	

Auto Indexing

Given the shown initial states, what is the value of R0 and R1 after executing LDR R1, [R0,#4]!

A) R0= 0x00000100 R1=0x12345678

B) R0= 0x00000104 R1=0x33221100

C) R0= 0x00000104 R1=0xAABBCCDD

D) R0= 0x00000100 R1=0xAABBCCDD

Answer is C

R0 0x00000100

R1 0x12345678

Initial State

Address Memory

0x100 0x00

0x101 0x11

0x102 0x22

0x103 0x33

0x104 0xDD

0x105 0xCC

0x106 0xBB

0x107 0xAA

0x108